Game set-up and timing

One table of 8, two observers

- Sticky notes: red 🟥, green 🟩, blue 🟦 and yellow 🟨 pads (pink too, for the 1-2-4-All)
- 8 red notes written in advance: `S1` … `S8`
- Pens, scrap paper cut in quarters (the parcels)
- 3 boards, the role sheets (Compute backend: one per executor), the two cards on the last page
- 6 players: one of each role. 7 to 8: the Compute backend gets a second and a third executor.

TIME	STEP
3 min	**Explain the game**
5 min	**Read sheets, set up the table**
7 min	**Round 1, happy path**
2 + 7 min	**Reset, round 2, a surprise failure**
2 + 7 min	**Reset, round 3, new seats and new packer rule**
7 min	**Debrief**

The table

SIMULATION board

1 red per parcel

Pool	Done	Outputs

RECONSTRUCTION board

2 green per parcel

Pool	Done	Outputs

MERGE board

4 blue per parcel

Pool	Done	Outputs

Seed stack
🟥 S1 … S8, written in advance

Ready
parcels waiting for the Dispatcher

Backend seat
🟩🟦🟨 pads, pens; 1 to 3 executors

Bin
every used parcel, done or failed. Never reused.

A parcel is a quarter sheet with the input notes stuck on it (`P5 · RECO`). After the job: ✓ inputs go to Done, ✗ inputs go back to their Pool, the empty parcel goes to the Bin.

A note's name says where it comes from: `G3` is the green file made from seed `S3` ; `B1 = G3 + G5` .

Simulation

You play the Simulation transformation's two plugins: its **feeder** and its **packer**. Each transformation has its own.

You hold: the seed stack S1 … S8, paper quarters, a pen.

1. **Feed:** put seeds 🟥 from the stack into your Pool, two or three at a time.
2. When the stack is empty, say aloud: **“Simulation feed is over.”**
3. **Pack:** 1 seed per parcel. Write `P1 · SIM` (then P2…) on a paper quarter, stick the seed on it, put it on **Ready**.

Reconstruction

You play the Reconstruction transformation's feeder and packer. Your feeder is an **edge feeder**: its inputs are the outputs of Simulation.

You hold: paper quarters, a pen. Your packer rule may change in round 3.

1. **Feed:** move each green note from the Simulation Outputs into your Pool.
2. When Simulation's feed is over, its **Done holds all 8 seeds** and its Outputs is empty, say: **“Reconstruction feed is over.”**
3. **Pack:** 2 green per parcel (`P7 · RECO`), on Ready. Not enough notes? Wait. Once your feed is over, pack what is left.

Merge

You play the Merge transformation's feeder and packer. Your feeder is an **edge feeder**: its inputs are the outputs of Reconstruction.

You hold: paper quarters, a pen.

1. **Feed:** move each  blue note from the Reconstruction Outputs into your Pool.
2. When Reconstruction's feed is over, its **Done holds all 8 green notes** and its Outputs is empty, say: **“Merge feed is over.”**
3. **Pack:** 4 blue  per parcel (`P12 · MERGE`), on Ready. Not enough notes? Wait. Once your feed is over, pack what is left.

Dispatcher

In the real system: the part that hands each parcel to a backend. There can be several backends; the dispatcher chooses one.

You hold: nothing.

1. Take the oldest parcel on **Ready**.
2. Give it to an executor of the Compute backend that is **free**: no parcel in its hands.
3. If every executor is busy, wait with the parcel. Never give an executor two parcels.

Compute backend

In the real system: where jobs run. All three transformations send their jobs here; the boards only keep the books. You are one **executor**: one parcel at a time.

You share: the  pads, pens. If the backend gets a card, read it in secret.

1. **SIM** parcel: for seed `S3`, write a green 🟩 `G3` and stick it on the parcel.
2. **RECO** parcel: for two green notes, write one blue 🟦 `B1 = G3 + G5`.
3. **MERGE** parcel: for the blue notes, write one yellow 🟨 `M1 = B1 + B2 + B3 + B4`.
4. Write ✓ next to each input that worked (✗ if not), then give the parcel to the Completion.

Completion

In the real system: the step that reads the job's report and decides what happens to each input.

You hold: nothing.

1. ✓ input: put it in the **Done** column of its board. Put the new notes in **Outputs**.
2. ✗ input: put it back in the **pool** of its board. The Packer will make a new parcel.
3. Put the empty parcel in the **Bin**. A parcel is never reused.
4. When `M1` is in Merge's Outputs, every Pool and Ready are empty and nobody holds a parcel, shout **“Completed!”**

Simulation board

1 red per parcel · print on A4 landscape

Pool

waiting to be packed

In: Simulation (seeds), Completion (✗ inputs)
Out: Simulation, when packing

Done

processed by exactly one parcel

In: Completion only (✓ inputs)
Out: nobody

Outputs

green notes, for the next board

In: Completion only (new notes)
Out: Reconstruction takes the green notes

Reconstruction board

2 green per parcel · print on A4 landscape

Pool

waiting to be packed

In: Reconstruction (green notes), Completion (✗ inputs)
Out: Reconstruction, when packing

Done

processed by exactly one parcel

In: Completion only (✓ inputs)
Out: nobody

Outputs

blue notes, for the next board

In: Completion only (new notes)
Out: Merge takes the blue notes

Merge board

4 blue per parcel · print on A4 landscape

Pool

waiting to be packed

In: Merge (blue notes), Completion (✗ inputs)
Out: Merge, when packing

Done

processed by exactly one parcel

In: Completion only (✓ inputs)
Out: nobody

Outputs

the yellow `M1` : the finished product

In: Completion only (new notes)
`M1` stays here: the game is over

Cards

COMPUTE BACKEND · ROUND 2 · SECRET

A job fails

Parcel `P3 · SIM` fails: write ✗ next to its seed, make no green note, pass the parcel on as usual.

When that seed comes back on a new parcel, it works.

RECONSTRUCTION · ROUND 3

New packer version

3 green  per parcel, instead of 2.

Once your feed is over, pack what is left.